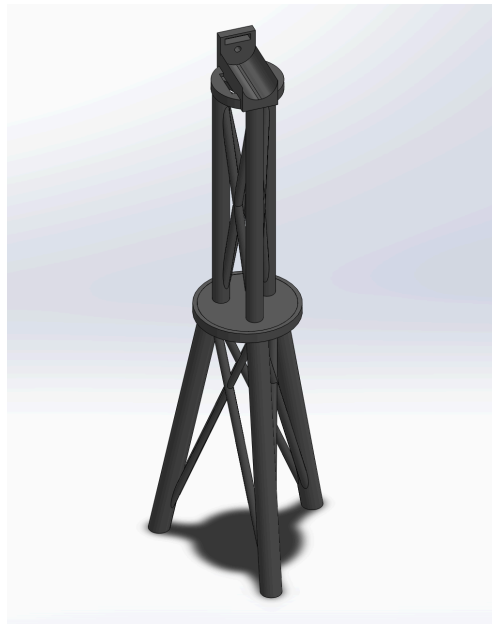




Wind Turbine Project Report

University of California, Berkeley

Mechanical Engineering Dept.

ENGIN 26, Group 21

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Professor Ken Youssefi

May 9, 2025

Project Summary

The wind turbine project involved the design, 3D modeling, fabrication, and testing of a functional wind turbine. The goal of this project was to model a scaled model of a wind turbine for renewable energy generation. In order to optimize the turbine's performance, it was crucial that all of the team members research and understand the theoretical concepts of its functionality. The three main components of the project were to design the rotor blades, support tower, and generator housing using SolidWorks. Once we printed all of the parts, the turbine was evaluated on two key criteria—power input and the structural stiffness of the tower.



Figure 1. Tower

In the initial stages of our project, our group first had to sketch and compile research on blade concept designs, finding optimal features that would efficiently generate maximum power while minimizing drag force and maximizing the amount of kinetic energy extracted from the wind. Our research consisted of finding measurements on airfoil shape, number of blades, angle of attack, and angle of twist. After finding ideal design features for our wind turbine blades, we then sketched our own tower structures based on existing designs that showed the most structural integrity, also taking into consideration weight and deflection. Once our group came into a consensus of each of the designs, we 3D modeled them, printed them ourselves, and assembled.

The last part of the project was testing the turbine based on power output of the blade and stiffness of the tower. From the data collected, our wind turbine produced a maximum power output of 1.332 W at 2.64V. The stiffness of the tower is calculated to be 19.62 N/mm at 0.1kg and 9.17 N/mm at 1kg. The wind turbine weighed 300g not including the base plate and the blades.

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Introduction

In an age of continually increasing demand for energy and ever dwindling natural resources, wind power, or power generated through wind energy rotating aerodynamic blades of a turbine, has become a widely popular renewable alternative to traditional coal and gas fuels.

Many wind turbines can be seen while driving on the highway past more rural areas as wind turbines need open space and strong winds to operate effectively. In 2023, 10% of the power consumed in the United States consisted of wind power (*Frequently Asked Questions (FAQs) - U.S. Energy Information Administration (EIA)*, n.d.). Our project is to design and create a wind turbine model that imitates the operations of a real one.

This project aims to challenge us in creativity and problem

solving; designing and engineering a wind turbine can be a lengthy

process to ensure its optimal performance. To ensure the safety of communities utilizing wind power, wind turbines must be built to withstand strong winds, especially because they are most effective in areas with high winds. The efficiency of wind turbines lies in the speed of the blades which, when connected to a generator, converts the kinetic energy and mechanical torque of the moving blades into electrical energy for people to use.



Figure 2. Wind Turbine

For maximizing the efficiency of the wind turbine, the features that must be optimized for the blade include the profile, number of blades, angle of twist, and angle of attack. The profile directly impacts how the turbine converts wind energy into rotational motion. The airfoil profile influences the balance between lift and drag where the optimal profile maximizes lift while minimizing drag which allows the blades to spin more efficiently and extract more energy from the wind (Wind Energy Technologies Office, 2023). Different profiles perform better at various wind speeds where thicker, rounder profiles are better suited for starting at low wind speeds, while thinner shapes excel at high efficiency operation in strong winds. Figures X-Z show some of our freehand sketches of the blade profile. In addition to optimizing the blade profile, the number of blades on a wind turbine significantly affects aerodynamic efficiency and structural behavior.

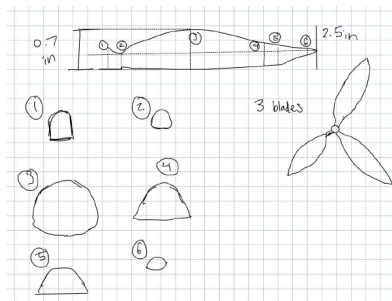


Figure X. Turbine Sketch 1

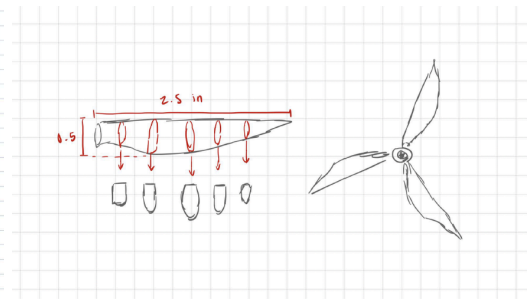


Figure Y Turbine Sketch 2

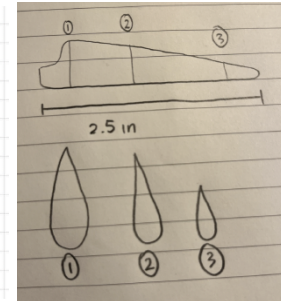


Figure Z Turbine Sketch 3

Studies have shown that the optimal number of blades for a wind turbine is 3 blades because it provides the most balance and stability to prevent equipment failure. As stated by Professor Youssefi, he mentioned that “a rotor with an even number of blades will cause stability problems for a wind turbine” producing an uneven force on the shaft and blade of the rotor. An even number of blades causes stability problems where the moment from the upper blade is reflected back and achieves its maximum power from the wind, and the lower blade passes through the

wind shade in front of the tower (Adeyeye et al., 2021). In addition to these features, we must also consider the angle of attack and angle of twist. The angle of attack must be optimized for the best lift to drag ratio. The angle of twist determines the maximum power efficiency across different wind speeds allowing for maximum torque and minimum drag.

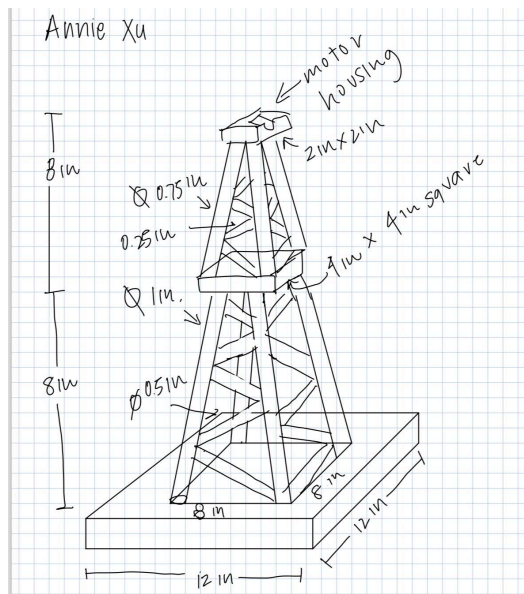


Figure 3. Tower Concept 1

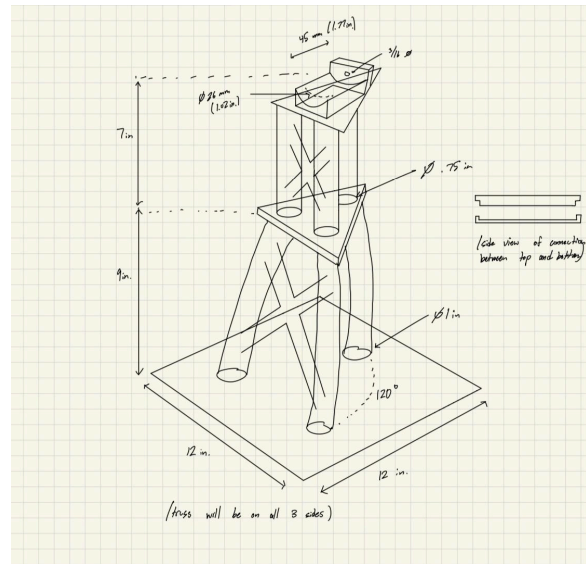


Figure 4. Tower Concept 2

Our design objectives were to optimize aerodynamic design of the turbine blades for higher power generation performance and maximizing structural integrity in terms of stiffness while reducing weight. Specifically, we wanted to create a wind turbine design which would have minimal deflection under a 1 kilogram load and generate a 2 watt power output while maintaining the weight of the 3D printed model under 300 grams. In addition, one of our requirements was to include a motor housing in our tower design that would securely hold the motor with room for zip ties. Many of our individual designs from the brainstorming process utilized trusses as well to ensure the print would not be over the target weight while still providing strong structural stiffness.

Design

Brainstorming:

When considering the blade designs for the wind turbine, there are a few things that must be considered. These include the blade profile, the angle of attack, the blade twist, and the number of blades. Of these three prototypes, the one on the top does not meet the specifications as it extends beyond the provided hub leaving our group to decide between the other two. We ended up choosing the design on the bottom as it was thinner and lighter, potentially providing the most power of the three blades.

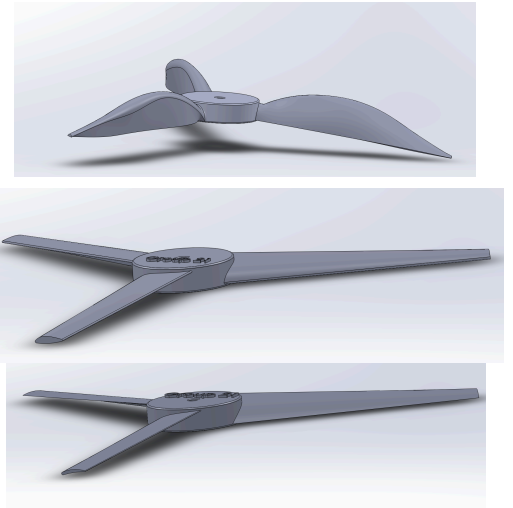


Figure 5. Blade Concepts

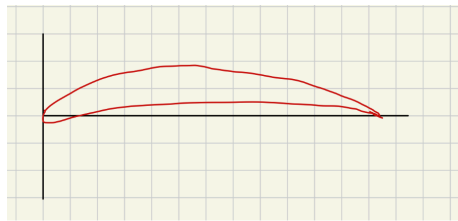


Figure 6. NACA 6409 Cross Section

This blade utilized the NACA 6409 airfoil shape. According to a study on optimal airfoil designs for small horizontal-axis wind turbines (Batu et al., 2024), the NACA 6409 was able to produce a significant amount of power. The study recommended that this airfoil should be used for power output and efficiency. Furthermore this design utilized 3 blades. For optimal efficiency 1 blade would perform the best however, there is an imbalance as a result. With more blades, there would be additional torque but also slower rotational speed due to increased drag and resistance (Adeyeye et al., 2021). Thus 3 blades is an efficient compromise between more and fewer blades.

The angle of attack and angle of twist that ended up on the final blade design were 8 degrees and 20 degrees respectively. The angle of attack of around 8 degrees should help to ensure the greatest amount of lift before the blades stall (Thumthae & Chitsomboon, 2009). An angle of twist is beneficial to ensure that the optimal angle of attack is achieved throughout the length of the blade. For smaller blades, an angle of twist of about 0 - 20 degrees from root to tip was found to be optimal as the twist does not have to be as extreme due to a smaller span (A.R. et al., 2016).

For tower designs, our main goal was to meet specifications with a volume beneath 17 in³, height from base to motor shaft of 16 in, and staying within the 12x12 sizing of the base. The final design ended up being a combination of the two sketches from the introduction.

The formula for the amount of power that is available from the wind is $P_{\text{wind}} = \frac{1}{2} \cdot \rho \cdot A \cdot V^3$ where ρ = air density, A = swept area of the turbine blades (m^2) = πR^2 , and V = wind speed (m/s). The measured power will not be the same as the potential power that the wind can generate as the turbine will not be able to capture all of the wind energy due to some being lost to heat/friction. Thus once we have the measurement from the tests we will be able to calculate efficiency which is given by $P_{\text{turbine}}/P_{\text{wind}}$ (Archer, 2021).

Software:

Solidworks was the software utilized to convert the hand-drawn sketches to 3D models. The design process involved creating sketches and using features such as lofts and extrusions to model the blades and components for the tower. The parts files for the tower could then be put together in an assembly file so we could use Finite Element Analysis (FEA) to simulate how the tower would react to real world testing.

Simulation:

To perform a simulation in Solidworks, it is critical to assign a material to the object. Our group printed the tower using PLA.

Unfortunately, Solidworks lacks a PLA preset so we had to manually input the material properties (data tables found in appendices) to create a custom material. A static simulation

was started after assigning material properties,

creating fixed points at the base of the tower and assigning a load to the back of the mount. In order to simulate using Solidworks, the software creates a mesh which essentially splits the component into many small parts to analyze how the component would react to external loads. If the mesh splits up the component into many parts, it has the potential to produce a more accurate simulation but it also takes longer to run the simulation. In figures 8, 9, and 10, the load was set to 1 kg or 9.81 Newtons of force with the default mesh setting.

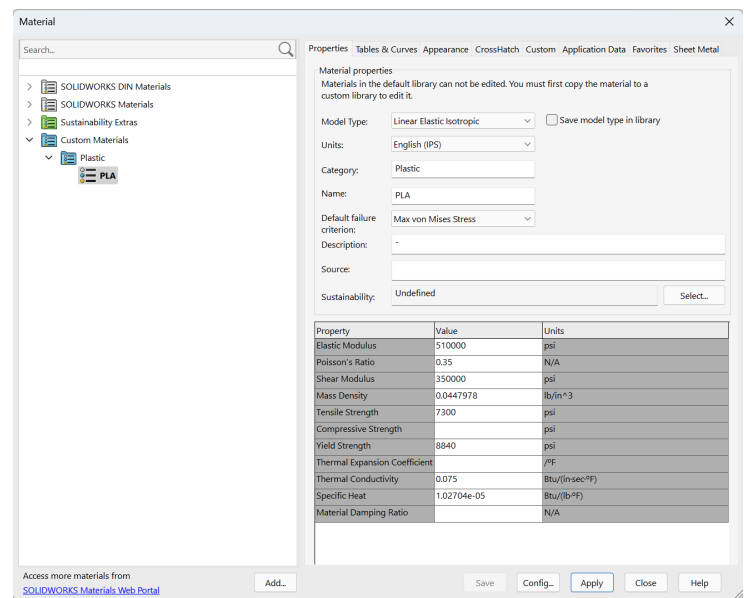


Figure 7. PLA Material Properties

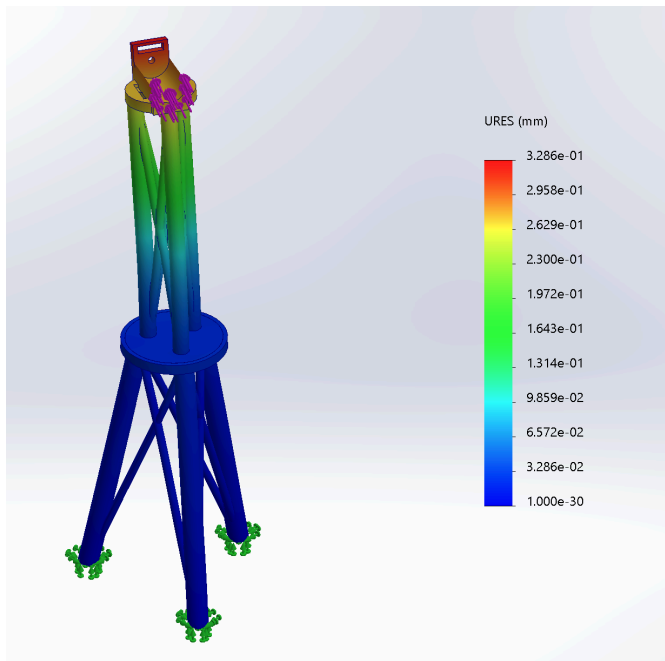


Figure 8. Displacement Simulation

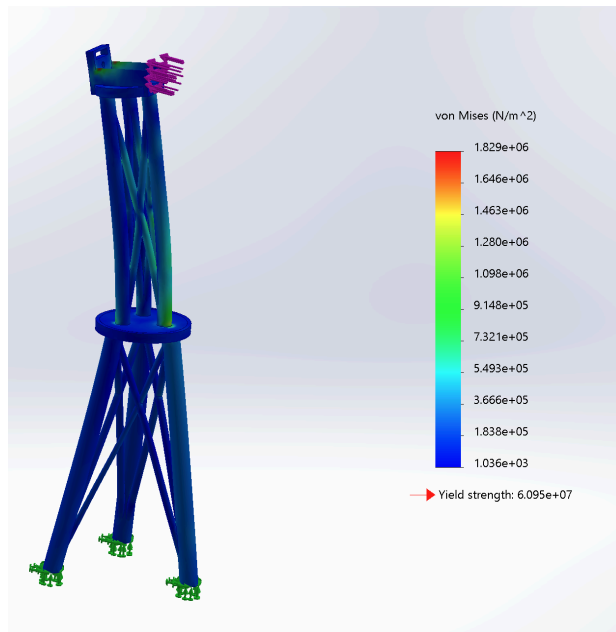


Figure 9. Stress Simulation

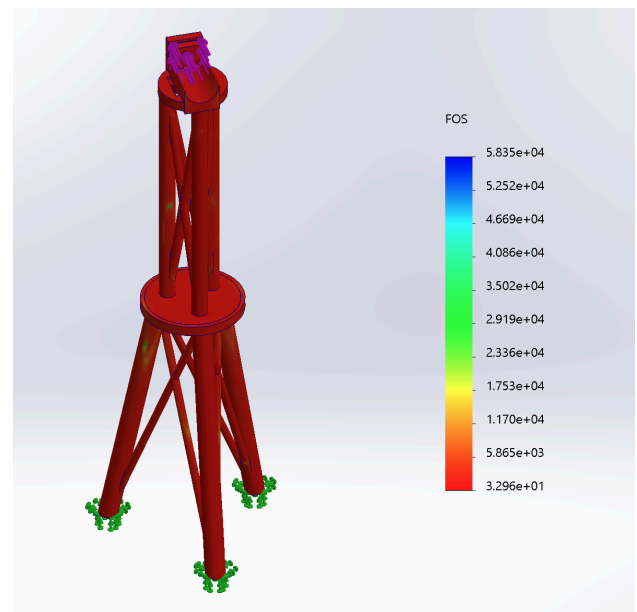


Figure 10. Factor of Safety

Simulation Results:

Figure 8. shows the total displacement/deflection that the tower faces under the load of 1 kg. From the simulation, one can observe that the maximum displacement occurs at the top of the tower, where the load was applied. From the color difference (blue to red), the deflection is shown to be 0 mm at the bottom to about 0.3286 mm at the top. This small displacement indicates the structure is very stiff under the load, suggesting the tower design is rigid and will be able to resist the force applied during testing.

Figure 9. demonstrates the von Mises stress on the tower under the load. The main area that shows a significant amount of stress is on the first vertical beam of the upper section of the tower which is around $1.463 \times 10^6 \text{ N/m}^2$. Additionally, there is stress on the mount where the motor housing connects to the back plate which holds the eye-bolt of around $1.829 \times 10^6 \text{ N/m}^2$. Both of these figures are below the yield strength of the material so there is not much to be concerned about for testing but there could be modifications made to the design for future work.

Figure 10. displays the factor of safety plot for the tower. Most of the tower in this simulation is red meaning it has a factor of safety of about 32.96. Thus, according to the simulation the tower may be able to handle about 33 times more than the load of 1 kg that was simulated (Safety Culture, 2024). The tower is safe and likely will not break when loaded with 1 kg of weight for real world testing.

Simulation Data and Graph

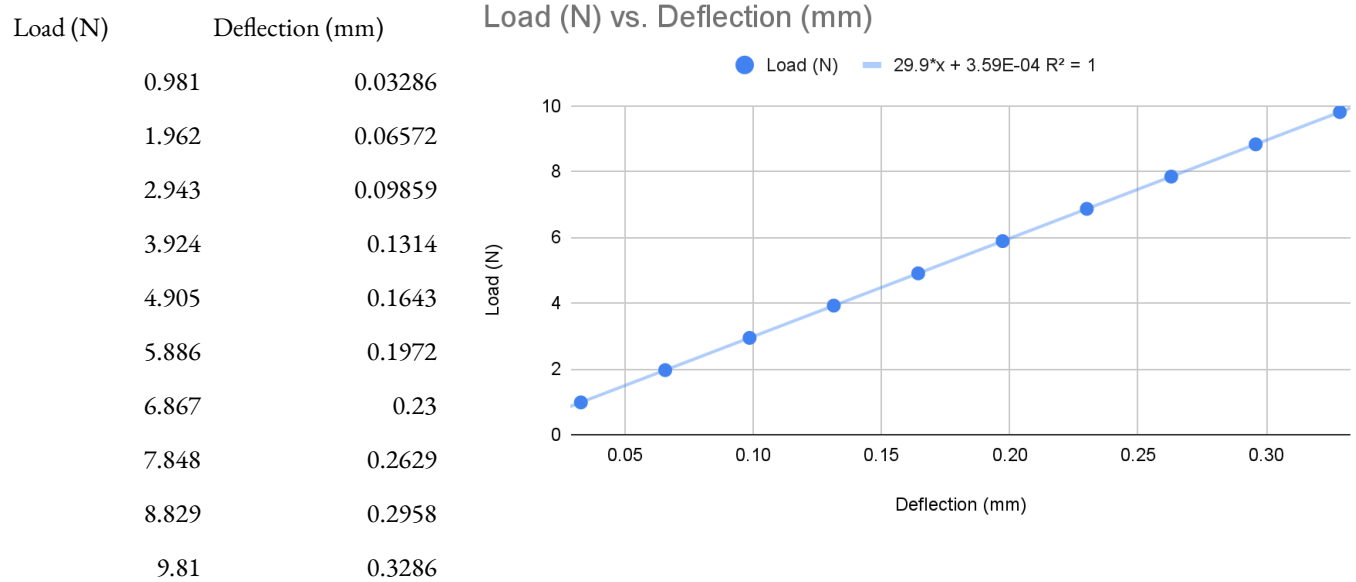


Figure 11. Simulation Load vs Deflection Graph

Volume:

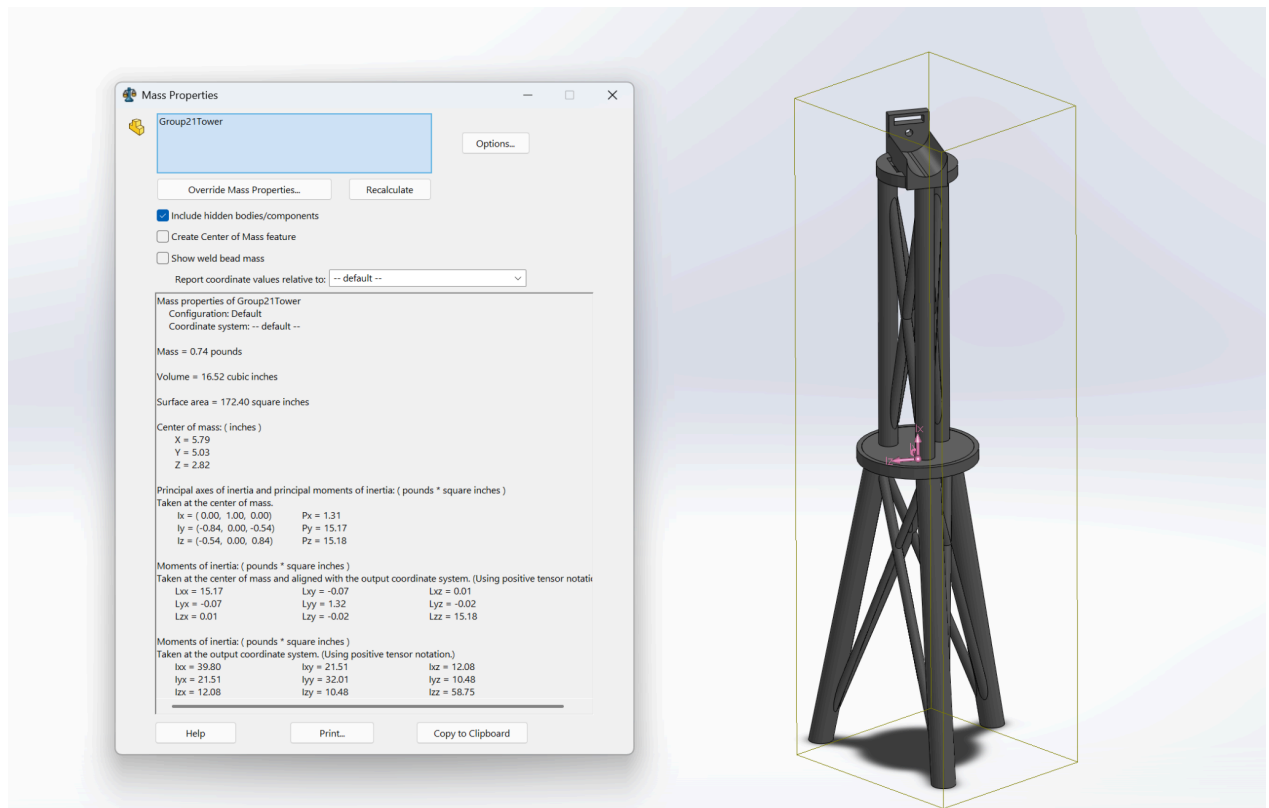


Figure 12. Tower Volume

Testing and Results

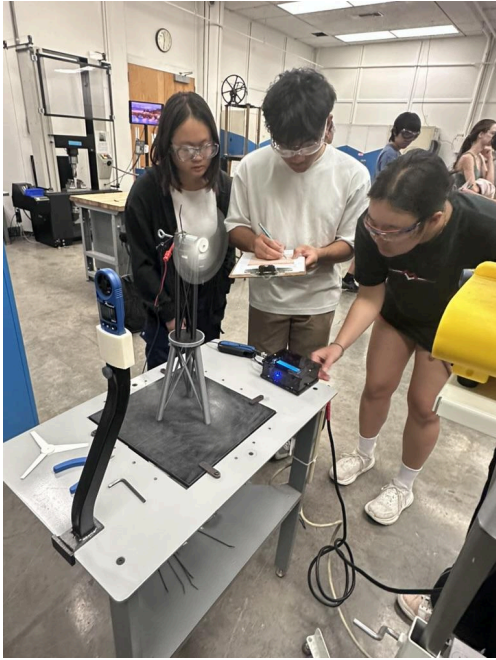


Figure 13. Power Generation Test



Figure 14. Tower Stiffness Test

Equipment Used:

Power Testing

- Digital Anemometer: device used to measure wind speed
- Power Meter (Load Box): device used to measure voltage, power, and current output based off of changing resistance
- Tachometer: device used to measure rotational speed (rpm) of the turbine
- Reflective tape: put on one blade so tachometer can produce output
- Wind Blower: blows wind to spin turbine blades

Stiffness Testing

- Eyebolt: screwed into mount to hook onto load
- Dial indicator: measures deflection
- Pulley with hooks: holds string that hooks onto load and tower
- Spacer: put in motor housing to connect dial indicator to tower

Testing Process:

Power testing was conducted with a constant wind speed. We first mounted the motor to the motor housing using a zip tie. Then we connected positive and negative leads from the power meter to the motor. To measure the wind speed from the wind blower, we used a digital anemometer and found that the wind speed was 25.0 mph. Once we removed the anemometer out of the way, the turbine started to spin allowing us to measure voltage, current, and power using the power meter. An optical tachometer and reflective tape was used to measure the blade rotational speed. After finding a measurement, we changed the resistance that the load box provided using the potentiometers in the box. This was done in order to optimize the power output that the blade could produce. The process of changing resistances was repeated 12 times to find the data points.

Stiffness testing was conducted by attaching a string to an eyebolt aligned with a pulley at the top of the tower. We first needed to align the dial indicator with a spacer placed in the motor housing. Then weight was added to the string in 100 g increments up to a total of 1 kg. A dial indicator fixed into place by a magnetic base was used to measure the displacement in millimeters as the weights are added.

Data Tables:Power Generation

Windspeed: 25.0 mph

Data Points	Voltage (V)	Current (mAmp)	Power (mWatt)	Blade Rotational Speed (rpm)
1	0.04	243.5	10.47	704.8
2	1.40	607.5	870.3	3409
3	2.26	560.6	1241	4453
4	1.90	578.0	1115	401.6
5	2.60	480.7	1252	446.7
6	2.64	481.3	1332	4747.9
7	2.82	344.1	994.1	4466
8	3.12	280.8	849.7	5135
9	3.32	222.1	732.5	5300
10	3.40	143.0	478.6	5461
11	3.51	150.1	440.0	5547
12	3.62	87.6	321.4	5733

Stiffness Generation

Data Points	Load (g)	Load (N)	Displacement (mm)	Stiffness (N/mm)
1	100	0.981	0.05	19.62
2	200	1.962	0.14	14.01
3	300	2.943	0.24	12.26
4	400	3.924	0.36	10.9
5	500	4.905	0.47	10.44
6	600	5.886	0.59	9.98
7	700	6.867	0.70	9.81
8	800	7.848	0.81	9.69
9	900	8.829	0.95	9.29
10	1000	9.81	1.07	9.17

Plots:

Load (N) vs Displacement (mm):

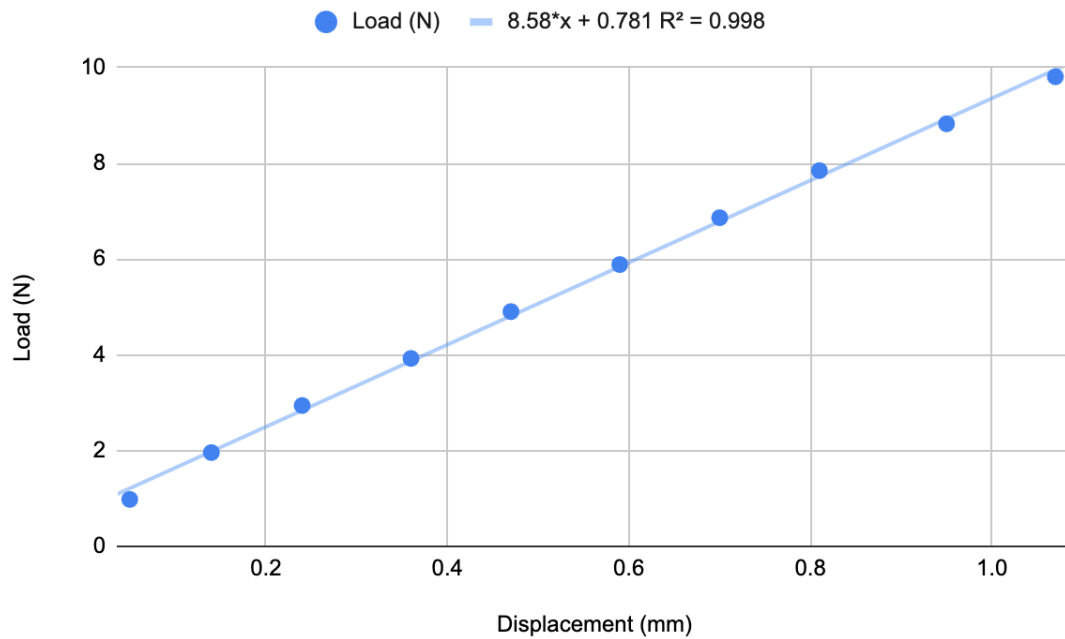


Figure 15. Measured Load vs. Displacement Graph

Power (mW) vs Current (mA):

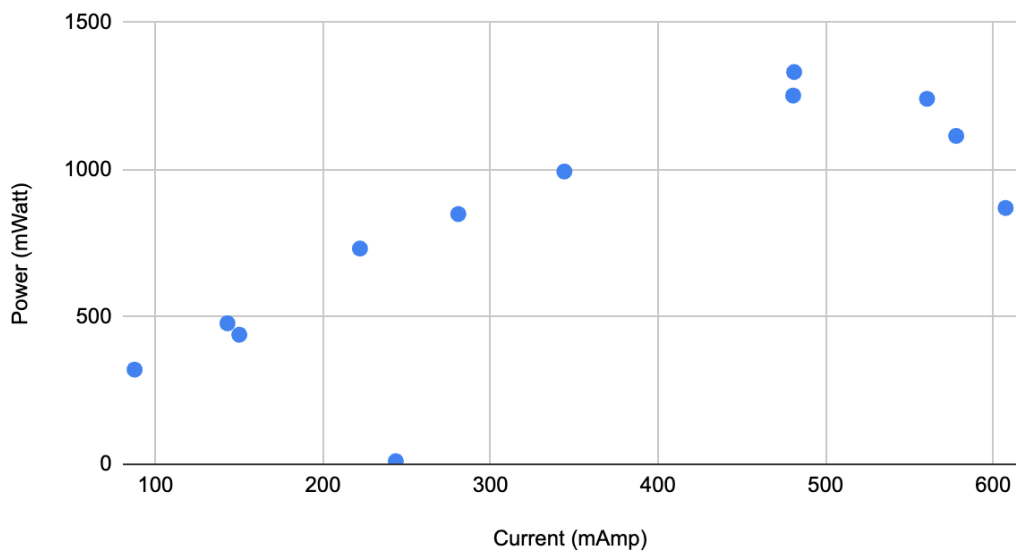


Figure 16. Measured Power vs. Current Graph

Voltage (V) vs Current (mA):

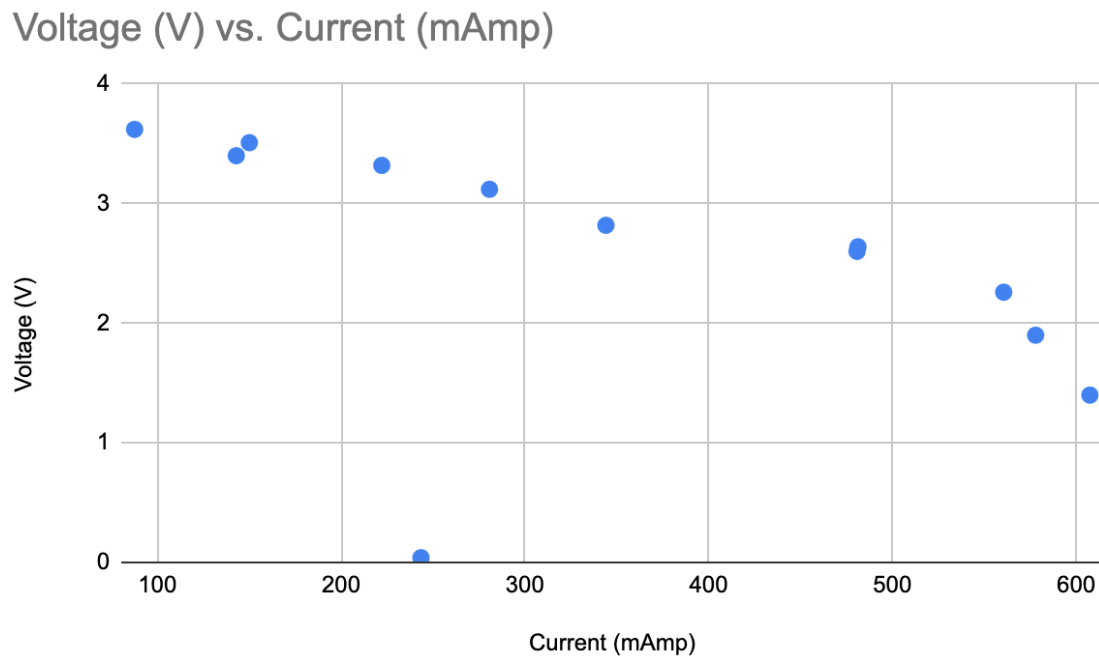


Figure 17. Measured Voltage vs. Current Graph

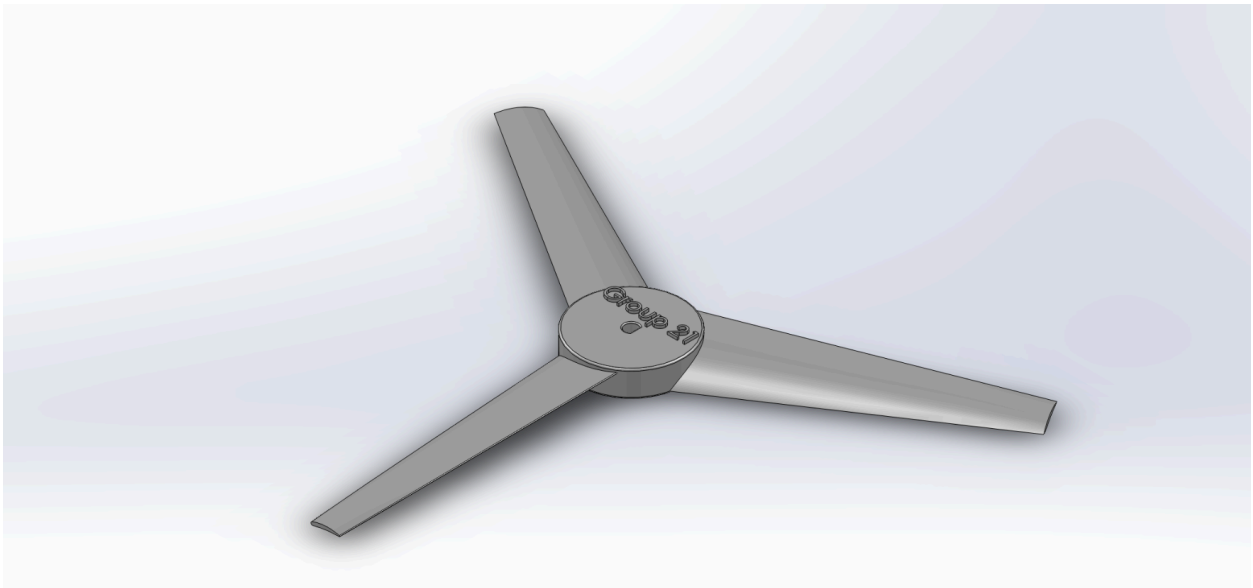
Efficiency:

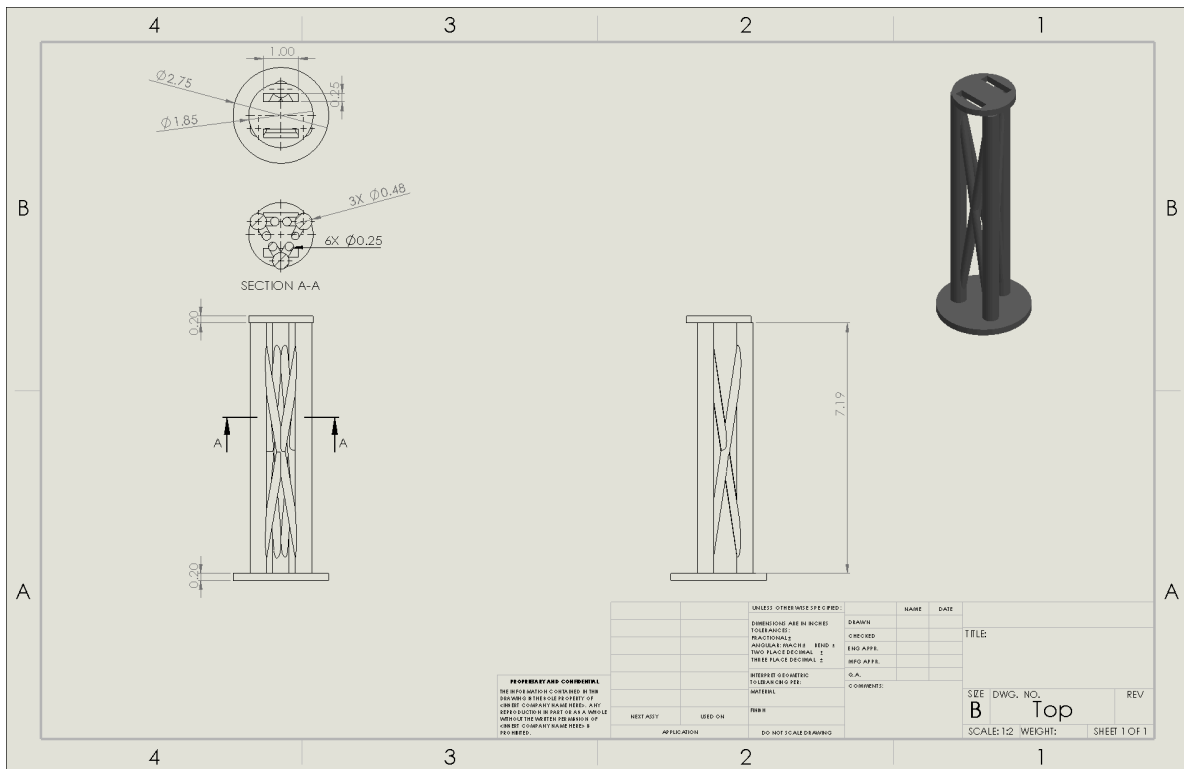
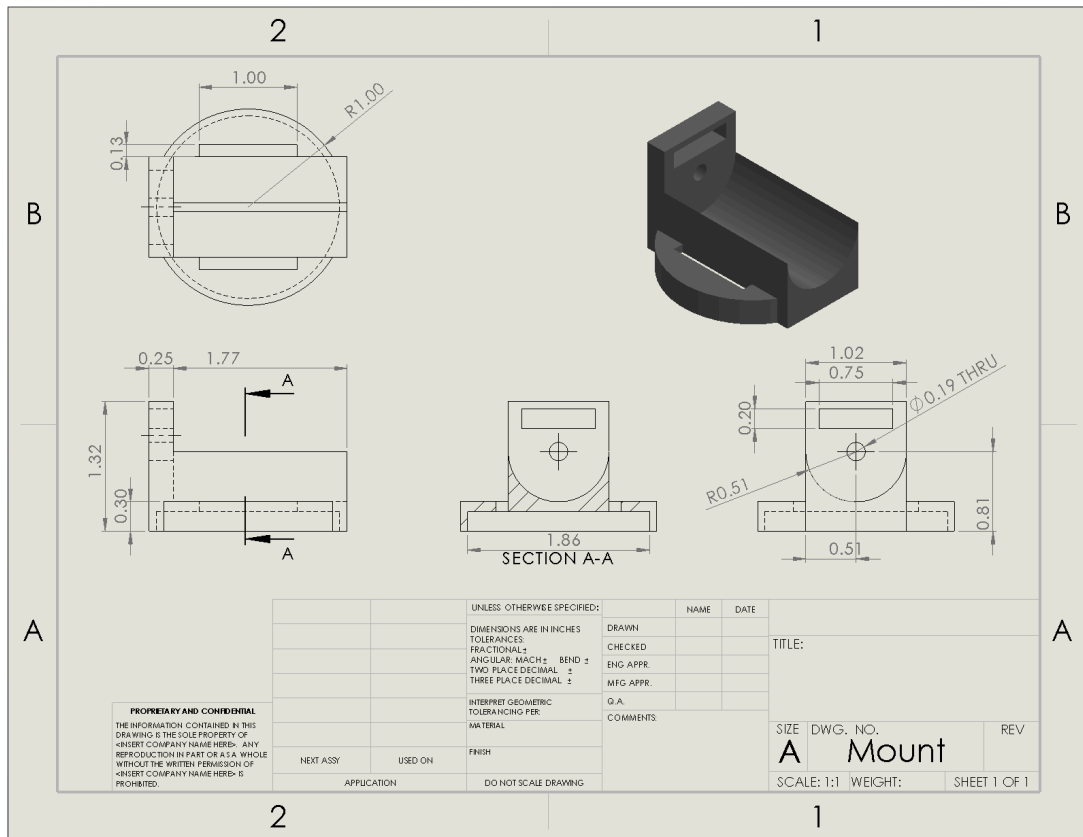
$$\text{Theoretical power} = \frac{1}{2} \cdot \rho \cdot A \cdot v^3 = \frac{1}{2} \cdot 1.225 \cdot 0.0182 \cdot (11.176)^3 = 15.56\text{W}$$

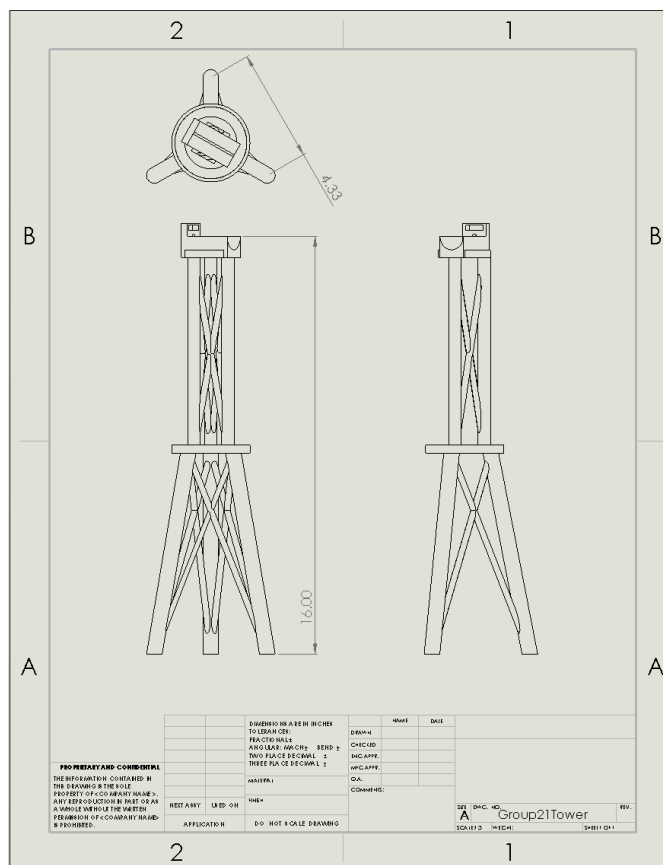
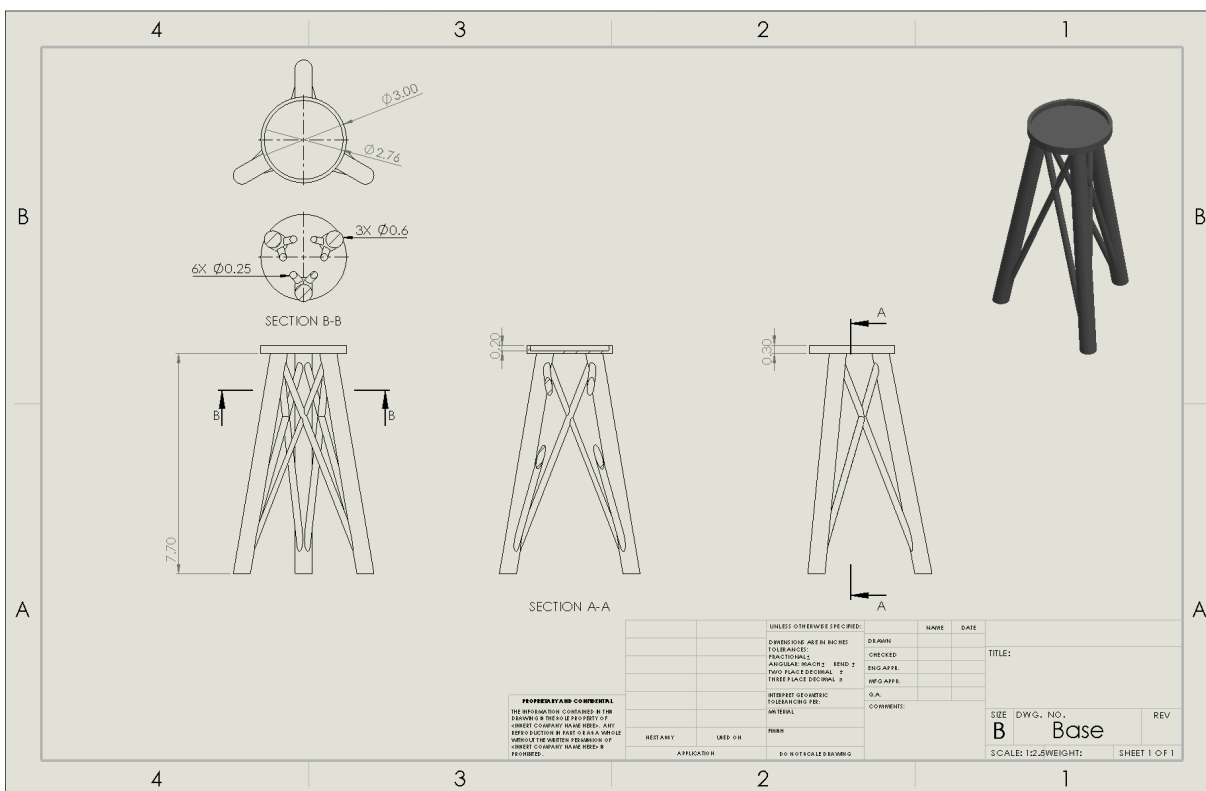
$$\text{Bertz: } 0.593 \cdot 15.56\text{W} = 9229 \text{ mW}$$

$$\text{Theoretical power/experimental power} = 9229 \text{ mW}/1332 \text{ mW} = 6.93$$

CAD Drawings







Conclusion

In this project, our group researched how wind turbines operated and their design choices—specifically, blade angle of attack, angle of twist, tower deflection. We then individually designed the blade and tower components of the wind turbine utilizing our research findings and collaboratively selected what we believed was the optimal design. To manufacture our wind turbine, we used a combination of Prusa and Bambu 3D printers with PLA filament to print our blade and the two sliced pieces of our tower before glueing with a strong adhesive. We tested the performance of our wind turbine design through power generation and stiffness testing at Hesse Hall. The highest power recorded was 1.332 watts with a controlled wind speed of 25 miles per hour and the displacement of the tower after a load of 1 kilogram was 1.07 millimeters. Overall, our design performed well but did not reach our goal of generating 2 watts of power. In addition, our wind turbine tower design did well for the stiffness test. It performed worse than the simulations but still achieved a stiffness of greater than 8. The lack in performance relative to the simulations is likely due to FDM printing where the PLA is connected via many layers rather than one solid piece, contributing to a more flexible product. Throughout this project, we learned to design with intention by spending time to research specific details, such as if the blades should be flat or curved, and going even further by printing multiple blades and testing their performance. It also provided us with technical experience in CAD, 3D printing, FEA simulations and physical performance testing of prototypes to optimize designs for specific objectives.

Recommendations

After seeing the results of the tower's performance, there are definitely ways to improve in the future or if we had more time. First of all, the body of our tower is on the heavier side. One way this could be revised in terms of design is to make the cross braces of the upper tower piece thinner. Since the upper piece of the tower is smaller than the lower piece in general, the supporting rods don't need to be as thick. In terms of manufacturing, the tower could have been made lighter by printing with less infill percentage. However, there is give and take with that because we want to maintain the stiffness of the tower while reducing weight.

For future work, our blade design can also be modified by reducing more of the blade width progressively towards the tip. Our current blade has a slight change in width across the blade but can be improved to generate power more efficiently because most power is generated near the outer third of the blade. By reducing material and drag at the tip, it will bring focus to the lift and generate more power. Reducing material at the tip of the blade by making it more tapered also reduces vibration, which allows the blade to rotate more smoothly and more stable, with less energy lost to friction.

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Wind Energy Technologies Office. (2023, May 10). Airfoils, Where the Turbine Meets the Wind.
Energy.gov. <https://www.energy.gov/eere/wind/articles/airfoils-where-turbine-meets-wind>

Appendices

Raw Data Sheets:

University of California
Mechanical Engineering Department – E26

Wind Turbine Power Generation test

Team member names: Sandy L., Alexander R., Michael W., Annie X., Hannah Z.

Lab section 102 Wind speed (mph) 25.0
Measured at the rotor location

Power Measurements

Data Points	Voltage (V)	Current (mAmp)	Power (mWatt)	Blade Rotational Speed (rpm)
1	0.04	243.5	10.47	304.8
2	1.40	604.5	840.3	340.8
3	2.26	560.6	1241	445.3
4	1.90	578	1115	401.6
5	2.6	470.7	1252	476.7
6	2.64	401.3	1332	474.9
7	2.82	341.1	994.1	496.6
8	3.12	280.8	849.7	513.5
9	3.32	227.1	732.5	530.0
10	3.40	143	478.6	546.1
11	3.51	130.1	446	554.7
12	3.62	87.6	321.4	573.3

University of California
Mechanical Engineering Department – E26

Wind Turbine Tower stiffness test

Team member names: Sandy L., Alexander R., Michael W., Annie X., Hannah Z.

Lab section Wednesday 12-2 Total weight (grams) 1195.0 g
- 895.1 g
300 g

Deflection Measurements

Data Points	Load (N)	Displacement (mm)	Observation
1	100g	0.05	
2	200g	.14	
3	300g	.24	
4	400g	.36	
5	500	.47	
6	600	.59	
7	700	.70	
8	800	.81	
9	900	.95	
10	1000	1.07	
11			
12			

PLA Specifications for Finite Element Analysis (FEA):

<https://www.makeitfrom.com/material-properties/Polylactic-Acid-PLA-Polylactide>

<https://www.seas3d.com/MaterialTDS-PLA.pdf>